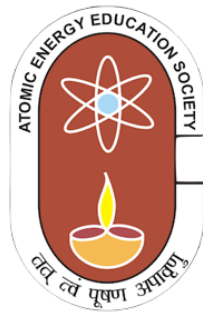


TRAJECTORY OPTIMISATION FOR A STEERABLE AIRDROP IN UNKNOWN WIND

Guiding Packages to a Target for Relief Distribution,
with No Wind Sensor on Board

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Interactive version: <https://1101-hub.github.io/guided-airdrop/>

Abstract

Relief supplies airdropped to flood-isolated communities drift with the wind and land away from the intended point. This work simulates a self-steering parafoil carrying no wind sensor and asks what bounds its accuracy. Dubins path planning is combined with a law producing a path of prescribed rather than minimum length; an Extended Kalman Filter (EKF) estimates wind, airspeed and heading bias from satellite velocity. Over $N = 150$ randomised drops the median miss distance falls from 8.57 m without a wind estimate to 3.89 m with the EKF (95% CI 2.85–5.28), a paired median reduction of 4.26 m. Turn radius scales linearly with wing loading and the loading needed to exceed wind speed W scales as W^2 , so the error floor scales as W^2 .

1 Introduction

A cloudburst on 19 July 2026 flooded four districts of Upper Assam; by 25 July, 705,000 people were affected and 80,000 remained unreachable (Tribune India, 2026; Onmanorama, 2026). Indian Air Force helicopters airdropped supplies, and the Chief Minister stated an intention to attempt “targeted drops in larger quantities” (ANI, 2026). A payload released without guidance is advected by the air mass throughout its descent: a requirement for targeted delivery implies current delivery is untargeted.

A parafoil is a ram-air canopy that glides, so horizontal displacement can be commanded during descent. Guided parafoils are established technology: the Naval Postgraduate School *Snowflake* has landed within 3 m of target (Naval Postgraduate School, n.d.), and the open-source ParaDrone plans Dubins paths on microcontroller hardware (Daniel, n.d.). This work claims no priority over either; it supplies a design analysis they do not report, relating wing loading and wind speed to the bound on landing error. Figure 1 shows one descent flown both ways.

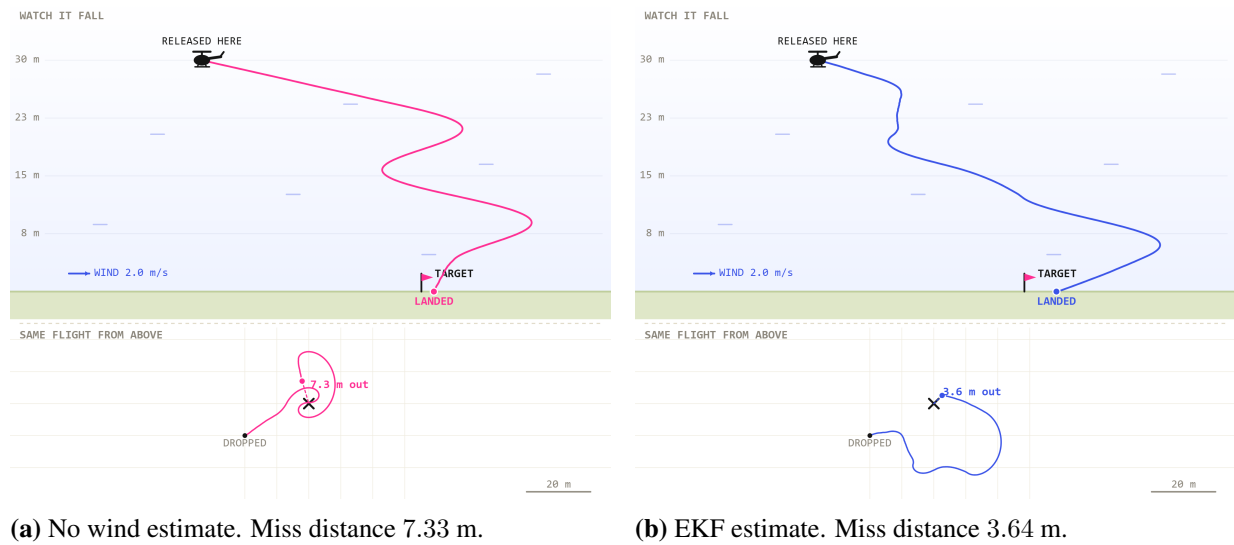


Figure 1: One descent, seen two ways, from the interactive model. The upper panel of each pair is the flight from the side, projected onto the release–target line, showing the canopy gliding forward as it descends at a constant rate; the lower panel is the same flight from above, where the miss distance is exact in both directions. The helicopter marks the release point, the flag and the cross mark the target. The two figures share identical axes and identical conditions — wind 2.0 m/s from the south, release height 30 m, wing loading 1.2 kg m^{-2} , magnetometer bias $+10^\circ$ — differing only in whether the guidance estimates the wind. These are single flights, shown to make the trajectory concrete; the statistical comparison is in table 1. The model can be flown interactively at <https://1101-hub.github.io/guided-airdrop/>.

2 Methods

Flight model. As a point mass in steady glide with lift equal to weight, the parafoil has airspeed

$$v = \sqrt{\frac{2g}{\rho C_L}} \sqrt{\frac{m}{S}}, \quad (1)$$

with ρ air density, C_L lift coefficient, m payload mass, S canopy area; only the wing loading m/S appears. When wind exceeds v the vehicle is advected downwind whatever the control input. Values used were $C_L = 0.80$, $C_D = 0.27$ and 20° bank.

Path planning. Under a minimum turn radius the shortest route between oriented configurations is a Dubins path: at most three turning or straight segments in one of six orderings (Dubins, 1957). All six closed forms were verified against step-wise integration (largest endpoint disagreement 9.0×10^{-12} m over 3,514 solutions).

Path-length control. Sink rate is constant, so time aloft and distance flown are fixed at release: the path needs prescribed, not minimum, length. A track at angle α to the bearing extends path length by $1/\cos \alpha$, giving the law $\cos \alpha = d/s$ for remaining distance d and budget s ; α alternates in sign. As altitude falls, $d/s \rightarrow 1$ and $\alpha \rightarrow 0$, giving a straight final approach with no mode change. Excess altitude is spent in a holding circle.

Wind estimation. No wind sensor is carried. An EKF (Kalman, 1960) maintains wind components w_x , w_y , airspeed v_a and magnetometer bias b from GNSS ground velocity, nonlinearity entering through b in the heading trigonometry. A headwind and an airspeed underestimate are indistinguishable on a fixed heading, leaving the state unobservable; turning separates them, and path-length control already turns.

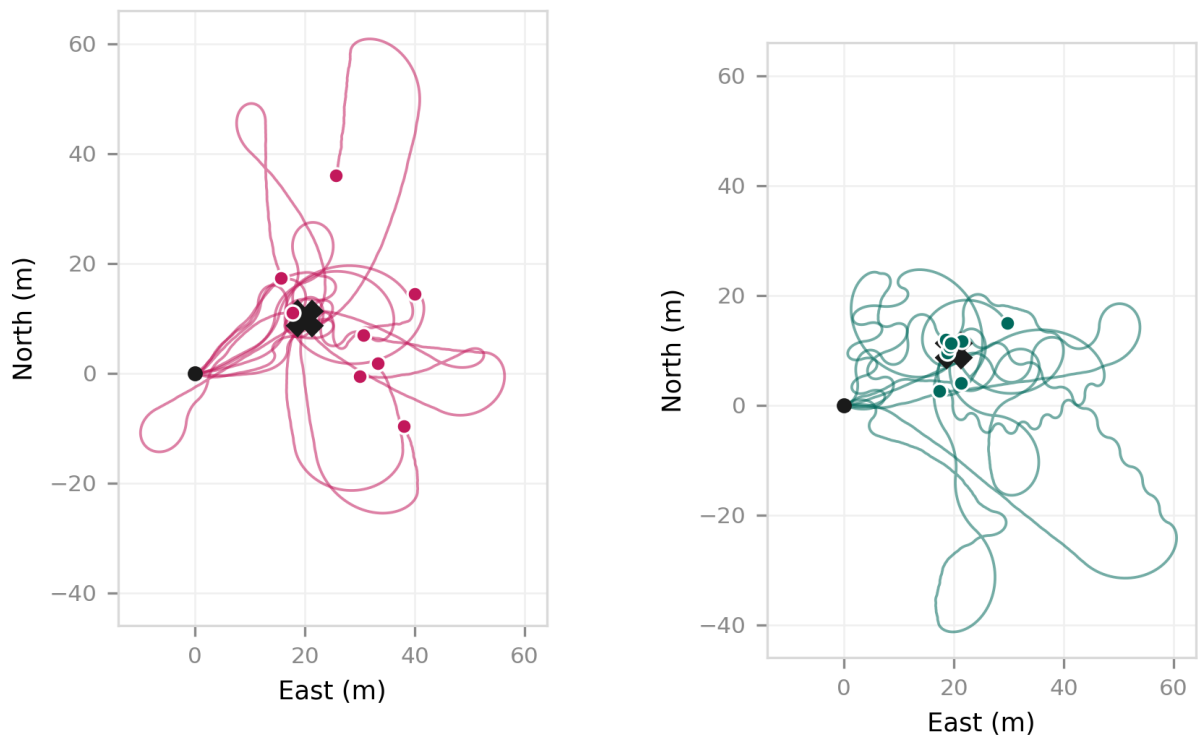
Simulation. Descent is integrated at $\Delta t = 0.05$ s, replanning every step, and guidance sees only GNSS velocity with Gaussian noise of 0.25 m s^{-1} and a magnetometer bias from $\pm 12^\circ$, never the simulated wind. Each trial draws wind $0\text{--}3.2 \text{ m s}^{-1}$ with uniform heading and release height $20\text{--}60$ m, flown three times under identical conditions: *no-estimate* (still air), *EKF* (estimated onboard) and *true-wind* (supplied). *Miss distance* is horizontal separation between touchdown and target; intervals are bootstrap percentiles from 20,000 resamples.

3 Results

Figure 2 shows eight descents per configuration; table 1 covers $N = 150$ randomised drops. Compared trial by trial, the median reduction from adding the EKF is 4.26 m (95% CI $3.08\text{--}6.57$), and the EKF gives the smaller miss distance in 112 of 150 trials; wind velocity is recovered to a median error of 0.104 m s^{-1} ($0.096\text{--}0.112$). The EKF and true-wind intervals overlap over most of their range, so those configurations are not separated at this sample size.

Table 1: Miss distance by guidance configuration over $N = 150$ randomised drops. Conditions are identical across rows and only the wind information available to the guidance differs. Intervals on the median are 95% bootstrap percentile intervals; the uncertainty on the mean is the standard error.

Configuration	Median (m)	Mean (m)	90th percentile (m)
No-estimate	8.57 (6.49–12.46)	11.52 ± 0.81	24.27
EKF	3.89 (2.85–5.28)	4.94 ± 0.31	10.75
True-wind	4.62 (3.37–5.44)	4.84 ± 0.32	10.93



(a) No-estimate configuration. Median miss distance 15.03 m.

(b) EKF configuration. Median miss distance 2.26 m.

Figure 2: Ground tracks of $N = 8$ descents, plotted in a target-centred frame with east and north in metres on identical axes in both panels. The same eight sets of conditions are flown in each panel, with wind speed drawn from $1.0\text{--}3.0\text{ m s}^{-1}$ and release height from $30\text{--}55\text{ m}$. The filled circle at the origin marks release, the cross marks the target, and filled circles at track ends mark touchdown. Touchdown points are dispersed in fig. 2a and concentrated near the target in fig. 2b.

Two scaling relations follow from eq. (1). Turn radius $R = v^2/(g \tan \phi)$ at bank angle ϕ , with $v \propto \sqrt{m/S}$, gives $R \propto m/S$: across loadings 0.3–2.4 kg m⁻² the ratio $R/(m/S)$ is 4.7694 m³ kg⁻¹, constant to $1.9 \times 10^{-14}\%$. Wind penetration requires $v > W$, giving $(m/S)_{\min} \propto W^2$: for $W = 2.0$ – 4.0 m s⁻¹ the ratio $(m/S)_{\min}/W^2$ is 0.058722 kg m⁻²(m s⁻¹)⁻², constant to $1.2 \times 10^{-14}\%$. Both sit at double-precision level: algebraic identities, not fits. Composing them gives

$$R_{\min} \propto W^2. \quad (2)$$

Figure 3 maps miss distance over this space.

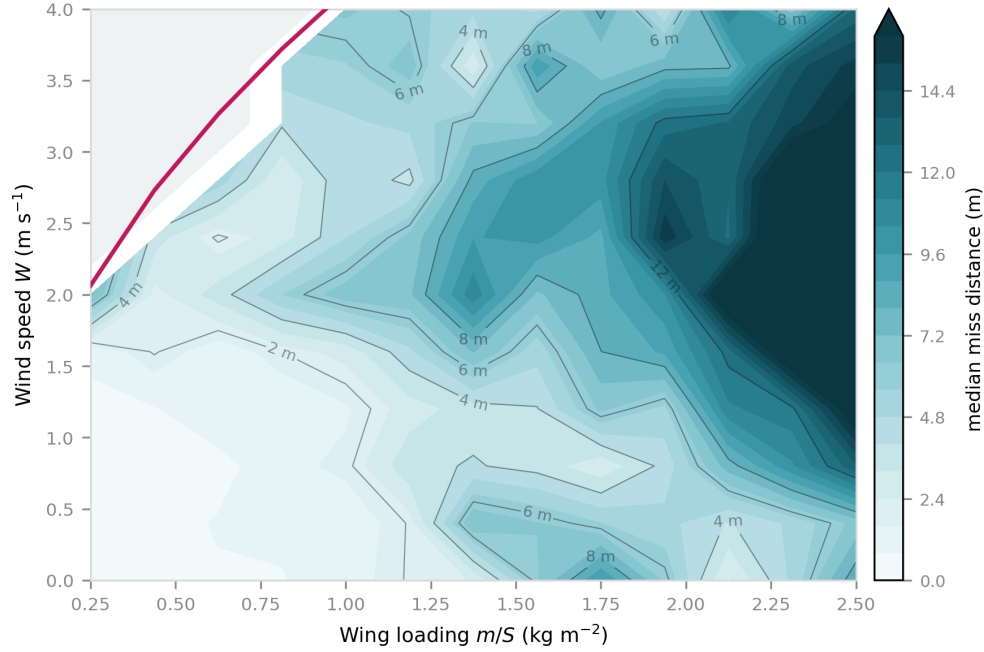


Figure 3: Median miss distance over wing loading m/S and wind speed W , from 24 drops at each of 13×11 grid points in the EKF configuration at a release height of 30 m. Contours are labelled in metres. The line marks $v = W$; in the shaded region below and left of it airspeed does not exceed wind speed and no guided descent is possible. Miss distance is smallest at low wing loading and low wind speed, and exceeds 12 m at the highest loadings irrespective of wind speed.

4 Discussion

The paired reduction establishes that onboard wind estimation improves accuracy here. That the EKF is not separated from true-wind is consistent with its correcting two error sources where true-wind corrects one: supplying wind leaves the $\pm 12^\circ$ magnetometer bias uncorrected, which the filter estimates jointly. Separating them would need a larger sample.

Equation (2) follows from two competing requirements: penetrating wind of speed W demands a wing loading rising as W^2 , while turn radius rises linearly with loading and bounds the smallest correction achievable in the final seconds. Specifying for wind never encountered therefore degrades accuracy in all conditions, including calm. The appropriate choice is the lowest loading satisfying $v > W$ with margin; set by ballast, it can be matched to the forecast before release, unlike canopy area.

Three limitations bound these results: turbulence is absent, so the wind field is uniform and steady and the accuracy reported is an upper bound; canopy deformation is excluded, fixing C_L and C_D ; and actuator dynamics are excluded, so turns begin instantaneously. Quantifying this optimism requires flight measurement; the model specifies a test configuration — 1 m² canopy, 0.5–0.8 kg payload, 20 m minimum release height — so it can be falsified rather than fitted.

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